M. VINITH KUMAR

COMPUTER SCIENCE & ENGINEERING UNDERGRADUATE

Full-Stack Development | IoT & Embedded Systems | Databases

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PROFESSIONAL SUMMARY

Computer Science and Engineering undergraduate (CGPA 8.0/10) with hands-on experience in full-stack web development, relational and document databases, and embedded IoT systems. Built an automated, solar-powered sensor-and-relay prototype and completed MongoDB certifications in vector search, retrieval-augmented generation (RAG) and AI agents. Seeking a software engineering internship or entry-level role.

TECHNICAL SKILLS

Programming Languages	Python, Java, C, C++, JavaScript
Web Development	HTML5, CSS3, JavaScript, React
Databases & Data Modeling	MySQL, MongoDB, Vector Search, Relational-to-Document Modeling
Embedded Systems & IoT	ESP32, Arduino, Sensors, Relays
Developer Tools	Git, GitHub, Docker, VS Code, Eclipse
Fundamentals	Data Structures & Algorithms, OOP, Database Systems, SDLC

PROJECT

GrainDrain: Automated Solar-Powered Grain Drying System — **IoT & EMBEDDED SYSTEMS**

- Designed and built a microcontroller-based system that monitors and regulates the microclimate around stored crops in real time.
- Programmed continuous temperature and humidity sensor readings to trigger relay-controlled cooling for crop preservation.
- Integrated solar power with energy-efficient control logic to reduce dependence on main-grid electricity.
- Tested system performance and data reliability under simulated agricultural drying conditions.

CERTIFICATIONS

MongoDB & AI Technologies Specialization

RAG with MongoDB	**Vector Search Fundamentals**	**AI Agents with MongoDB**
Relational to Document Model	**MongoDB Student Basics**	**Data Architecture**

EDUCATION

Bachelor of Engineering (B.E.), Computer Science & Engineering — **CGPA: 8.0 / 10.0**
Mount Zion College of Engineering and Technology, Pudukkottai

Higher Secondary Certificate (Class XII) — **77%**
Kamban Karpagam Matriculation Higher Secondary School, Karaikudi

Secondary School Leaving Certificate (Class X) — **71%**
Kamban Karpagam Matriculation Higher Secondary School, Karaikudi

AREAS OF INTEREST

Full-stack development | IoT and embedded automation | Data-driven systems | Vector search and AI applications